

Notes will be posted on my homepage:

<https://zhc515.github.io/teaching>

Reminder: Midterm 6:30-7:30 pm

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OH: Lunt B14, Mon, 2:30-4:00 pm

- (0) Curve
- (1) Limits and Continuity
- (2) Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$.
- (3) ∇f and $Du f$.
- (4) Tangent plane and linearization
- (5) Taylor's formula / Quadratic approximation

} practice
quiz 7

(3) $\vec{\nabla} f$ and Duf

$$-1 \leq \cos \theta \leq 1$$

● Assume $|\vec{u}|=1$, then

$$D_{\vec{u}} f(a, b) = \vec{\nabla} f(a, b) \cdot \vec{u} = |\vec{\nabla} f(a, b)| \cdot 1 \cdot \cos \theta \leq |\vec{\nabla} f|$$

where θ is the angle between $\vec{\nabla} f(a, b)$ and $\vec{u}$.

$D_{\vec{u}} f(a, b)$ maximum $\Leftrightarrow f$ increases most rapidly along $\vec{u}$

$$\Leftrightarrow \vec{u} \parallel \vec{\nabla} f(a, b)$$

$$\Leftrightarrow D_{\vec{u}} f(a, b) = |\vec{\nabla} f(a, b)|.$$

$D_{\vec{u}} f(a, b)$ minimum $\Leftrightarrow f$ decreases most rapidly along $\vec{u}$

$$\Leftrightarrow \vec{u} \parallel -\vec{\nabla} f(a, b)$$

$$\Leftrightarrow D_{\vec{u}} f(a, b) = -|\vec{\nabla} f(a, b)|.$$

(3) ∇f and Duf

(Quiz 7 #1)

In what direction is $f(x,y) = x^2 \cdot y^2$
increasing the fastest at $(1,2)$?

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

$$= \langle 2x \cdot y^2, x^2 \cdot 2y \rangle$$

$$\nabla f(1,2) = \langle 8, 4 \rangle$$

$$\vec{u} = \frac{\nabla f(1,2)}{|\nabla f(1,2)|} = \frac{\langle 8, 4 \rangle}{\sqrt{8^2 + 4^2}}.$$

In direction of $\vec{u}$, increasing
the fastest.

(4) Tangent plane and linearization

- Tangent plane to a level surface $F(x,y,z)=C$ at (a,b,c)

$$\vec{\nabla} F(a,b,c) \cdot (x-a, y-b, z-c) = 0$$

i.e. normal is $\vec{\nabla} F(a,b,c)$

- Tangent plane to the graph of $f=f(x,y)$ at (a,b,c) :
 $c=f(a,b)$

$$f_x(a,b) \cdot (x-a) + f_y(a,b) \cdot (y-b) - (z-c) = 0$$

i.e. normal is $\langle f_x(a,b), f_y(a,b), -1 \rangle$

$$z = f(x,y)$$

$$(=) \underbrace{f(x,y) - z}_{} = 0$$

$$\underbrace{F(x,y,z)}_{} = 0$$

(4) Tangent plane and linearization

(Quiz 7 #2)

Find an equation of the tangent plane to the **graph** of $f(x,y) = x^2 \cdot y^2$ when $(x,y) = (1,2)$

$$\begin{aligned} \frac{\partial f}{\partial x} &= 2x \cdot y^2, & \frac{\partial f}{\partial y} &= x^2 \cdot 2y & \therefore f(a,b) \\ \frac{\partial f}{\partial x}(1,2) &= 8 & \frac{\partial f}{\partial y}(1,2) &= 4 & c = f(1,2) \\ & & & & = 1^2 \cdot 2^2 = 4 \end{aligned}$$

$$\begin{aligned} \frac{\partial f}{\partial x}(1,2)(x-1) + \frac{\partial f}{\partial y}(1,2)(y-2) - (z-4) &= 0 \\ 8 \cdot (x-1) + 4 \cdot (y-2) - (z-4) &= 0 \end{aligned}$$

(Quiz 7 #3) $8x + 4y - z - 12 = 0.$

Find an equation of the tangent plane to the surface $x^2 + y - z^3 = 3$ at the point $(2,0,1)$.
 $= \bar{F}(x,y,z)$

$$\frac{\partial \bar{F}}{\partial x} = 2x \quad \frac{\partial \bar{F}}{\partial y} = 1 \quad \frac{\partial \bar{F}}{\partial z} = -3z^2.$$

$$F_x(2,0,1) = 2 \cdot 2 = 4 \quad F_y(2,0,1) = 1 \quad F_z = -3 \cdot 1^2 = -3$$

$$\begin{aligned} F_x(2,0,1)(x-2) + F_y(2,0,1)(y-0) + F_z(2,0,1)(z-1) &= 0 \\ 4 \cdot (x-2) + 1 \cdot (y-0) + (-3) \cdot (z-1) &= 0 \end{aligned}$$

$$4x + y - 3z = 5$$

(4) Tangent plane and linearization

● The linearization of $f(x,y)$ at (a,b) is
$$L(x,y) \stackrel{\text{def}}{=} f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b).$$

● Error estimate:

$$E(x,y) \stackrel{\text{def}}{=} f(x,y) - L(x,y) \quad \text{"tail of Taylor expansion"}$$

$$|E(x,y)| \leq \frac{1}{2} M (|x-a| + |y-b|)^2$$

where $|f_{xx}|, |f_{xy}|, |f_{yy}| \leq M$ around (a,b) .

(Quiz 7 #4)

Find the linear approximation of

$$f(x,y) = 2x^2 - 3y^2 \text{ at } (1,1),$$

and approximate the error if $|x-1| \leq 0.1$

and $|y-1| \leq 0.1$.

$$L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$$

$$a=1, \quad b=1$$

$$f_x(1,1) = 4x|_{(1,1)} = 4 \cdot 1 = 4$$

$$f_y(1,1) = -6y|_{(1,1)} = -6 \cdot 1 = -6$$

$$f(1,1) = 2 \cdot 1^2 - 3 \cdot 1^2 = -1.$$

$$\begin{aligned} L(x,y) &= -1 + 4 \cdot (x-1) + (-6) \cdot (y-1) \\ &= 4x - 6y + 1 \end{aligned}$$

$$f(x,y) = 2x^2 - 3y^2 \text{ at } (1,1)$$

$$f_{xx} = 4$$

$$f_{xy} = 0 = f_{yx} \quad \text{What is } M?$$

$$f_{yy} = -6$$

M is a number

$$\text{s.t. } |f_{xx}| \leq M$$

$$|f_{xy}| \leq M$$

$$|f_{yy}| \leq M$$

$\stackrel{\text{def}}{M} = 6$ (M can be larger than 6)

$$\begin{aligned} |E(x,y)| &\leq \frac{1}{2} M \cdot (|x-a| + |y-b|)^2 \\ &\leq \frac{1}{2} \cdot 6 \cdot (0.1 + 0.1)^2 \end{aligned}$$

$$\begin{aligned} &\text{if } |x-1| \leq 0.1 \\ &\text{and } |y-1| \leq 0.1 \end{aligned}$$

$$= 0.12$$

(4) Tangent plane and linearization (Summary)

- Tangent plane to a level surface $F(x,y,z)=C$ at (a,b,c) :

$$\vec{\nabla} F(a,b,c) \cdot (x-a, y-b, z-c) = 0$$

i.e. normal is $\vec{\nabla} F(a,b,c)$

- Tangent plane to the graph of $f=f(x,y)$ at (a,b,c) :
 $c=f(a,b)$

$$f_x(a,b) \cdot (x-a) + f_y(a,b) \cdot (y-b) - (z-c) = 0$$

i.e. normal is $\langle f_x(a,b), f_y(a,b), -1 \rangle$

- The linearization of $f(x,y)$ at (a,b) is
 $L(x,y) \stackrel{\text{def}}{=} f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$.

- Error estimate:

$$E(x,y) \stackrel{\text{def}}{=} f(x,y) - L(x,y)$$

$$|E(x,y)| \leq \frac{1}{2} M (|x-a| + |y-b|)^2$$

where $|f_{xx}|, |f_{xy}|, |f_{yy}| \leq M$ around (a,b) .

(5) Quadratic Approximation

● $f(a,b) + f_x(a,b) \cdot h + f_y(a,b) \cdot k + \frac{1}{2} [f_{xx}(a,b) \cdot h^2 + \overset{f_{xy} + f_{yx}}{2f_{xy}(a,b)} \cdot hk + f_{yy}(a,b) \cdot k^2]$, $(a+b)^2$
where $h = x - a$, $k = y - b$.

Quiz 7 # 5 Find the quadratic approximation of $f(x,y) = e^y \sin(x)$ at $(0,0)$ $f(0,0) = e^0 \cdot \sin 0 = 0$

$$f_x(0,0) = e^y \cdot \cos(x) \Big|_{(0,0)} = 1$$

$$f_y(0,0) = e^y \cdot \sin(x) \Big|_{(0,0)} = 0$$

$$f_{xx}(0,0) = \frac{\partial}{\partial x} (e^y \cdot \cos(x)) = e^y (-\sin(x)) \Big|_{(0,0)} = 0$$

$$f_{xy}(0,0) = \frac{\partial}{\partial y} (e^y \cdot \cos(x)) = e^y \cdot \cos(x) \Big|_{(0,0)} = 1$$

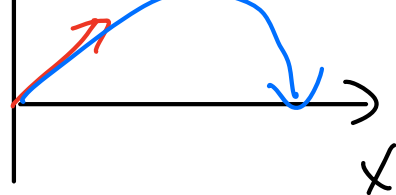
$$f_{yy}(0,0) = \frac{\partial}{\partial y} (e^y \cdot \sin(x)) = e^y \cdot \sin(x) \Big|_{(0,0)} = 0$$

$$\begin{aligned} Q(x,y) &= f(0,0) + f_x(0,0)(x-0) + \cancel{f_y(0,0)(y-0)} \\ &+ \frac{1}{2} (\cancel{f_{xx}(0,0)} \cdot (x-0)^2 + 2f_{xy}(0,0)(x-0)(y-0) + \cancel{f_{yy}(0,0)} \cdot (y-0)^2) \\ &= f(0,0) + 1 \cdot (x-0) + \frac{1}{2} \cdot 2 \cdot 1 \cdot (x-0)(y-0) \\ &= 0 + x + xy = x + xy. \end{aligned}$$

$$Q(x,y) = x + xy$$

$$Q(x,y) = x + x \cdot y$$

(0) Curve
(22W)



5. (8 points) A tennis ball is launched inside of a wind tunnel. It is launched from the point $(0, 0, 0)$ with an initial speed of 50 m/s in the direction of the vector $\langle 0, 4, 3 \rangle$ and its trajectory is affected by gravity and the wind such that its acceleration in m/s^2 is given by

$$\mathbf{a}(t) = \langle -3, -1, -10 \rangle.$$

Determine the distance in meters between the launch point of the ball and the point at which it lands. (Give an exact numerical answer, but you do not need to simplify).

Trajectory: $\mathbf{a}(t) \xrightarrow[\mathbf{v}(0)]{\text{integrate}} \mathbf{v}(t) \xrightarrow[\mathbf{r}(0)]{\text{integrate}} \mathbf{r}(t)$

$$\mathbf{r}(0) = \langle 0, 0, 0 \rangle$$

$$\mathbf{v}(0) = 50 \cdot \frac{\langle 0, 4, 3 \rangle}{\sqrt{0^2 + 4^2 + 3^2}} = \langle 0, 40, 30 \rangle$$

$$\mathbf{a}(t) = \langle -3, -1, -10 \rangle$$

$$\begin{aligned} \mathbf{v}(t) &= \int \mathbf{a}(t) dt = \langle \int -3 dt, \int -1 dt, \int -10 dt \rangle \\ &= \langle -3t + A, -t + B, -10t + C \rangle \end{aligned}$$

$$\mathbf{v}(0) = \langle 0, 40, 30 \rangle \Rightarrow A = 0, B = 40, C = 30$$

$$\mathbf{v}(t) = \langle -3t, -t + 40, -10t + 30 \rangle$$

$$\vec{r}(t) = \int \vec{v}(t) dt$$

$$= \left\langle \int -3t dt, \int (-t + 40) dt, \int (-10t + 30) dt \right\rangle$$

$$= \left\langle -\frac{3}{2}t^2 + A, -\frac{1}{2}t^2 + 40t + B, -5t^2 + 30t + C \right\rangle$$

$$\vec{r}(0) = \langle 0, 0, 0 \rangle \Rightarrow A = B = C = 0$$

$$\Rightarrow \vec{r}(t) = \left\langle -\frac{3}{2}t^2, -\frac{1}{2}t^2 + 40t, -5t^2 + 30t \right\rangle.$$

$$-5t^2 + 30t = 0 \Rightarrow t = 6$$

$$t = 6,$$

$$\vec{r}(6) = \left\langle -\frac{3}{2} \cdot 6^2, -\frac{1}{2} \cdot 6^2 + 40 \cdot 6, -5 \cdot 6^2 + 30 \cdot 6 \right\rangle$$

$$= \langle -54, 222, 0 \rangle$$

$$|\vec{r}(6) - \vec{r}(0)| = \sqrt{54^2 + 222^2}.$$

(0) Curve

(22 W)

2. Let $\mathcal{C}$ be the curve parametrized for $t \geq 0$ by

$$\mathbf{r}(t) = \langle \sin t - t \cos t, \cos t + t \sin t \rangle; \quad t \geq 0$$

(a) (6 points) Find the arc length of $\mathcal{C}$ between the points $(0, 1)$ and $(\pi, -1)$.

$$\begin{aligned} \mathbf{r}'(t) &= \langle \sin t' - (t' \cdot \cos t + t \cdot \cos t'), \cos t' + (t' \cdot \sin t + t \cdot \sin t') \rangle \\ &= \langle t \sin t, t \cos t \rangle \end{aligned}$$

$$|\mathbf{r}'(t)| = \sqrt{(t \sin t)^2 + (t \cos t)^2} =$$

Determine a , and b in $\int_a^b |\mathbf{r}'(t)| dt$: $a=0, b=\pi$

$$\text{Length} = \int_0^\pi |\mathbf{r}'(t)| dt = \int_0^\pi t dt = \frac{1}{2} \pi^2.$$

Formula: $\text{Length} = \int_a^b |\vec{r}'(t)| dt$

(0) Curve

(b) (4 points) Find an arc length parametrization of $\mathcal{C}$ beginning at the point $\mathbf{r}(0)$.

Arc length parametrization:

1) Compute $S(t) = \int_0^t |\vec{r}'(\tau)| d\tau$

2) Write $t = t(s)$

3) Plug $t = t(s)$ back to $\vec{r}(t)$.

1) Length between time 0 and t is

$$S(t) = \int_0^t |\vec{r}'(\tau)| d\tau = \int_0^t \tau d\tau = \frac{1}{2}t^2$$

2) Describe t by $s \Rightarrow t = \sqrt{2s}$

3) Plug $t = \sqrt{2s}$ back to $\vec{r}(t) = \langle \sin t - t \cos t, \cos t + t \sin t \rangle$.

$$= \langle \sin \sqrt{2s} - \sqrt{2s} \cos \sqrt{2s}, \cos \sqrt{2s} + \sqrt{2s} \sin \sqrt{2s} \rangle$$

(1) Limits and Continuity

For "0/0" limit, two path test, factorization,
Finally, try $x = r \cos \theta$, $y = r \sin \theta$ (very rare!)

(23 W) Compute the following limits, or show that they do not exist.

(b) (4 points) $\lim_{(x,y) \rightarrow (1,1)} \frac{x - \sqrt{y}}{x^2 - y}$

(23 S) Compute the following limits, or show that they do not exist.

(c) (4 points)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 y^4}{x^{12} + y^6}$$

(1) Limits and Continuity

For "0/0" limit, two path test, factorization,
Finally, try $x = r \cos \theta$, $y = r \sin \theta$ (very rare!)

(22 W) Compute the following limits, or show that they do not exist.

(b) (6 points)

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 y^2}{x^2 + y^2}$$

(1) Limits and Continuity

$$f(x,y) \text{ is continuous at } (a,b) \Leftrightarrow \lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$$

(23W)

6. Let $f(x,y) = \begin{cases} 1, & y = 0 \\ 2y + 1, & x = 0 \\ x + 2y, & \text{elsewhere} \end{cases}$

(a) (4 points) Prove that $f(x,y)$ is not continuous at $(0,0)$ (hint: use limits).

(24W) True or false

(b) (4 points) The function

$$f(x,y) = \begin{cases} \frac{x^4 - y^4}{x^2 + y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0) \end{cases}$$

is continuous at $(0,0)$.

(2) Compute $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ (direct approach)
(23W)

1. Let $f(x, y) = e^{3x} \cos(3y)$

(a) (6 points) Calculate the partial derivatives f_x , f_y , and f_{xy} . Clearly box your final answers.

(b) (4 points) Show that $f_{xx} + f_{yy} = 0$.

(2) Compute $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ (Chain Rule)
(24W)

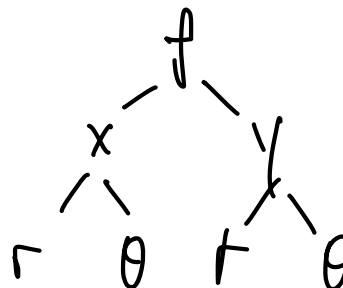
5. Let $f(x, y)$ be a function with continuous partial derivatives satisfying

$$f_x(1, 1) = -2, \quad f_y(1, 1) = 5, \quad f_x(\sqrt{2}, \pi/4) = 17, \quad f_y(\sqrt{2}, \pi/4) = -3.$$

(a) (12 points) Consider the polar coordinates

$$x = r \cos \theta, \quad y = r \sin \theta.$$

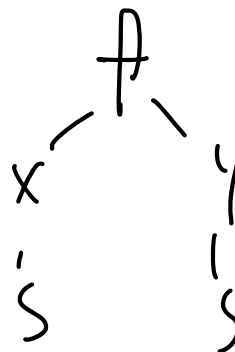
Find $\frac{\partial f}{\partial r}$ and $\frac{\partial f}{\partial \theta}$ at $(r, \theta) = (\sqrt{2}, \pi/4)$.



$$\begin{matrix} x & y \\ | & | \\ r & \theta \end{matrix}$$

(b) (8 points) Consider the parametrized curve $\mathbf{r}(s) = (\sqrt{2}s, \pi s^2/4)$. Compute

$$\left. \frac{d}{ds} f(\mathbf{r}(s)) \right|_{s=1}.$$



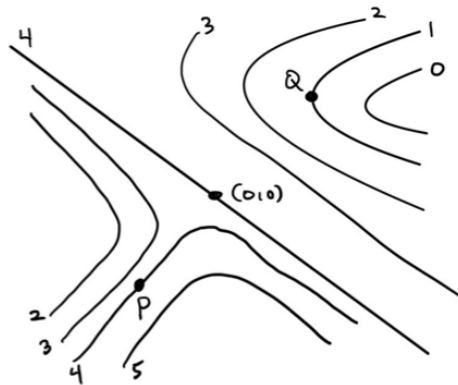
(3) $\vec{\nabla} f$ and Duf

$$f = f(x, y), \quad \vec{\nabla} f \perp \text{tangent line of level curve}$$

$$\vec{\nabla} f \text{ gives direction of fastest increasing of } f.$$

(23 F)

3. (This problem has two parts and continues on the next page.) Suppose $z = f(x, y)$ is a continuous function with nonnegative values and the following level curves:



The labels on the level curves are the values of z at which they occur. Assume that other level curves occur at values of z strictly between those that are drawn. The x -axis (not drawn) passes through $(0, 0)$ horizontally, and the y -axis (not drawn) passes through $(0, 0)$ vertically.

- (a) (10 points) Assuming that f_x and f_y exist and are continuous at all points, determine whether each of the following is positive, negative, or zero.

$$f_x(P), \quad f_y(P), \quad f_x(Q) - f_y(Q)$$

Justify your answer for $f_x(Q) - f_y(Q)$ only.

(3) ∇f and $D_u f$

$$D_u f(a,b) = \nabla f(a,b) \cdot \frac{\vec{u}}{|\vec{u}|}$$

(23 F)

5. Let $f(x,y) = \frac{3e^y}{x^2-y}$.

(a) (10 points) Find the rate at which f is changing at $(-3, 1)$ in the direction of $\frac{2}{\sqrt{13}}\mathbf{i} + \frac{3}{\sqrt{13}}\mathbf{j}$.

$$f(x,y) = \frac{3e^y}{x^2-y}$$

$$\frac{\partial f}{\partial x} = -\frac{3e^y \cdot 2x}{(x^2-y)^2} \quad \frac{\partial f}{\partial y} = \frac{3e^y \cdot (x^2-y) - 3e^y \cdot (-1)}{(x^2-y)^2}$$

$$\Rightarrow \nabla f(-3,1) = \left\langle \frac{-18e}{64}, \frac{-27e}{64} \right\rangle.$$

(b) (10 points) Find the direction in which the rate of change of f at $(-3, 1)$ is as negative as possible and the rate of change in this direction.

$$|\vec{u}| = 1 \quad D_u f = \nabla f \cdot \vec{u} = |\nabla f| \cdot \underbrace{|\vec{u}|}_{=1} \cdot \cos \theta$$
$$\Rightarrow \cos \theta = -1$$

Solutions of worksheet are posted on Canvas.